

ACTIVITY #11 Exploring Network Commands in Kali Linux

Task 1.1: Check IP Address

- **IP Address:** 10.0.2.15
- **Network Interface:** eth0

Task 1.2: Check Gateway

- **Default Gateway:** 10.0.2.2

Task 2.1: Ping a Website

- **Did the request succeed?:** Yes
- **What is the average time?:** 3.035 ms

Task 3.1: Convert Domain to IP

- **What IP address is returned?:**

(Two IP address was present)

- **IPv4:** 74.125.204.138
- **IPv6:** 2404:6800:4017:80c::200e

Task 3.2: Trace Route

- **How many hops does it take to reach Google?:** 30 hops max

Task 4: View Active Ports

- **List at least 2 open ports:**
- **What service is running on those ports?:**

(NO PORTS WERE GIVEN IN “SS-TULNP” COMMAND)

Task 5: Scan a Target:

- **What ports are open?:** Ports 80/443
- **What services are detected?:** HTTP and HTTPS

Task 6.1

- **What interface will you use?:** eth0

Task 6.2: Capture Traffic

- **What kind of traffic did you see?**
 - ARP traffic
 - DNS traffic
 - IPv6 traffic
 - IPv4 traffic
- **Did you observe any IP addresses?**
 - IPv4 (10.0.2.15 / 10.0.2.2 LOCAL & DEFAULT GATEWAY)
 - IPv6 (fe80::77a8:1777:c103:4a34)

PART 7: Focus Questions

- **Why do some commands require sudo?**
 - **Root Access:** protects system integrity by restricting critical system changes.
 - **Security Control:** prevents unauthorized users or malware from altering firewall rules or reading restricted system logs.
- **What is the importance of nmap in cybersecurity?**
 - to Identifies open ports and the specific software/services running on them and helps analysts spot outdated software, exposed entry points, and misconfigurations.
- **How can tcpdump help in monitoring attacks?**
 - because it inspects raw network traffic entering or leaving an interface in real time.